

Skytree Installation - Single Node

Release 15.4.0

SKYTREE®

Notices

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Customers with a support contact can contact support using any of the following methods:

- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

Freemium customers can send inquiries to the LinkedIn® Community Group for Skytree:

<https://www.linkedin.com/groups/4468542>

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Contents

Overview 1

Requirements 1

 Hardware Requirements 1

 Software Requirements 1

 Additional Assumptions 1

 Supported Browsers 1

Installed Packages 2

Running the Script 2

What's Next? 4

Overview

This document describes how to install Skytree on a single node using the `hadoopinstall.sh` script.

Requirements

Be sure that your system meets the following requirements before installing Skytree.

Hardware Requirements

- **Minimum:** Your hardware must have at least 16 physical cores with 128 GB of RAM (DDR3 or faster).
- **Recommended:** We recommend at least 40 physical cores with at least 320 GB of RAM (DDR3 or faster).

Software Requirements

The following are assumed to be already available on your system:

Operating System

The operating system must be one of the following:

- Red Hat® Enterprise Linux® (RHEL) 6.5
- CentOS™ 6.5

Python 2.7

The Skytree Python SDK requires Python 2.7.

In addition, Setuptools must already be bootstrapped. The Python SDK cannot install without Setuptools. Any version compatible with Python 2.7 is supported. Refer to pypi.python.org/pypi/setuptools for installation instructions.

Additional Assumptions

- The installation requires Internet access.
- The user must have root access on the machine that will host Skytree.
- Ports 8080 and 8088 must be accessible.

Supported Browsers

The table below lists the browser/OS combinations that are supported in Skytree. The supported browser version is assumed to be the current browser version.

Table 1: Supported Browsers

Browser	OS
Google Chrome™	Windows® 7, Windows 8, Mac OS X®
Mozilla Firefox®	Windows 7, Mac OS X

Installed Packages

The installation script installs the following packages onto your node:

- Apache™ Hadoop® 2.3 with Zookeeper™
- OpenJDK™ (Java Development Kit) 1.8
- Apache Tomcat™ 7.0.53
- Skytree Server™ 15.4.0
- Skytree Platform™ 15.4.0
- PostgreSQL® 9.4
- Spark™ version 1.5.1 with Scala 2.11
- Curl

Running the Script

Perform the following steps to run the Skytree single-node installation script.

1. Log on to <https://downloads.skytree.net/singlenode> and download the following file onto your machine:

- hadoopinstall.tar

This package includes the following files:

- catalina.properties
- context.xml
- core.xsl
- ._DS_Store
- .DS_Store
- hadoopinstall.sh
- hdfs.xsl
- mapred.xsl
- single-node-skytree-installer.sh
- skytree.properties
- ._spark-assembly-1.5.1-hadoop2.3.0.jar

- spark-assembly-1.5.1-hadoop2.3.0.jar
- tomcat.xml
- versions
- yarn.xml

2. Open a terminal window or command prompt and navigate to the location where you saved the downloaded file.
3. Unzip the hadoopinstall.tar file using the following command:

```
sudo tar -xf hadoopinstall.tar
```

4. Navigate to the newly created hadoopinstall folder, and launch the Apache Hadoop installer using the following command. This will install and then start Apache Hadoop.

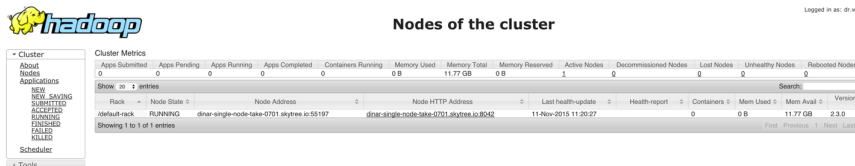
```
sudo sh hadoopinstall.sh
```

Note: The installer may prompt you to provide the Java version that you want to use. Be sure to select: `/opt/jdk1.8.0_65/bin/java`

The following message will display on the screen upon completion:

Hadoop successfully installed and started.

5. To verify that the installation was successful, open a supported browser. Navigate to `http://<fqdn>:8088` (replacing `<fqdn>` with the correct fully qualified domain name), and verify that Hadoop is running.



Nodes of the cluster

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
0	0	0	0	0	0 B	11.77 GB	0 B	1	0	0	0	0

Show: all 1 entries

Node	Node State	Node HTTP Address	Last health-update	Health-report	Containers	Mem Used	Mem Avail	Version
default-ec2-1-1011010	drn-r-single-node-late-0701-anything-1011010	drn-r-single-node-late-0701-anything-1011010	11-Nov-2019 11:20:37		0	0 B	11.77 GB	2.3.0

Showing 1 to 1 of 1 entries

6. When Hadoop is up and running, start the Skytree installer using the following command:

```
sudo sh single-node-skytree-installer.sh
```

7. Specify the path where you want to download the Skytree binaries. For example:

```
/home/ec2-user/hadoopinstall/skytree-binaries
```

After the Skytree software installation is complete, the script will start the Skytree platform and then run the smoke tests. The following output will display after the smoke tests complete successfully.

```
Running Skytree REST Api smoke tests for host URL http://localhost:8080
[TestNG] Running:
Command line suite
.
.
.
.
=====
Command line suite
Total tests run: 5, Failures: 0, Skips: 0
=====
Completed tests!
```

What's Next?

Now that you have successfully installed Skytree, you can begin preparing data and training and testing models through the REST API, Python SDK, or the Skytree GUI. Refer to the following documents for more information. These documents are available by logging on to <https://support.skytree.net>.

- *Skytree REST API Guide*
- *Skytree Python SDK Guide*
- *Skytree GUI Guide*